

Jaime Pesca

Optimization & Algorithms | Mathematical Modeling & Decision Support

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RESEARCH PROFILE

Industrial engineer with an M.S. in Applied Analytics (with distinction), specializing in the formulation of hard optimization problems and the design of algorithmic strategies to solve them. My work pairs methodological development with validation on real-world data and has produced one Scopus-indexed publication and two working papers. I aim to pursue doctoral research that advances the algorithmic and theoretical foundations of these methods while keeping them grounded in practical applications.

PUBLICATIONS & WORKING PAPERS

- **[Published]** da Silva-Ovando, A. C., Granados-Rivera, D., Pesca Santos, J. E., Mejia, G., & Mejia-Argueta, C. (2025). *A Multi-product Competitive Facility Location Model for Food Purchase: Does the Store Type Matter?* In Sustainable Systems in an Internetworked Digital World (Proc. ICPR 2025), Lecture Notes in Production Engineering, Springer. Peer-reviewed, Scopus and EI Compendex indexed.
- **[In revision]** Pesca, J., Mejia, G., & Gutierrez-Franco, E. *Locating Street Markets using Qualitative and Quantitative Decision Models* (MINLP and Fuzzy AHP). Presented at the INFORMS / ALIO / ASOCIO conference 2024. Revising for resubmission following peer review.
- **[In preparation]** Pesca, J., Gutierrez-Franco, E., & Mejia-Argueta, C. *Optimizing Pallet Configuration for Air Transport of Cut Flowers*. Online 3D bin packing optimization validated on real charter-flight data. Presented at the MIT Global SCALE Network 2026.
- **[In revision]** Pesca, J., & Ysa K. *The Cycle-Constrained Aerial Suppression Base and Water Point Location Problem: Joint Siting under Uncertainty with an Application to Cundinamarca, Colombia*.

TECHNICAL & METHODOLOGICAL SKILLS

- **Optimization:** MILP, MINLP, large-scale combinatorial optimization, constraint modeling, multi-stage / hierarchical formulations, heuristics and metaheuristics, model benchmarking and runtime analysis.
- **Statistics & Decision Methods:** discrete choice / random utility modeling, Fuzzy AHP and multi-criteria decision analysis, design of experiments and A/B testing, Machine Learning, Deep Learning.
- **Programming & Tools:** Python, R, SQL, Gurobi, GAMS, Git.
- **Languages:** English (C1, IELTS overall 7.0), Spanish (native).

EDUCATION

Graduate Certificate in Logistics and Supply Chain Management

May 2025 to Feb 2026

Massachusetts Institute of Technology (MIT), Cambridge, MA, USA

- Final grade: 87 / 100. Awarded a 50% merit scholarship.

M.S. in Applied Analytics

Jun 2023 to Feb 2026

Universidad de La Sabana, Bogota, Colombia

- Grade average: 4.58 / 5.0 (distinction-level). Full merit research fellowship (100%).

Professional Certificate in Data Science and Analytics

Apr 2023 to Nov 2023

Massachusetts Institute of Technology (MIT), Online

B.S. in Industrial Engineering

Jan 2018 to Jun 2022

Universidad de La Sabana, Bogota, Colombia

- Grade average: 3.84 / 5.0. Top national State Exam (Saber Pro) scores in Industrial Engineering.

RESEARCH & TEACHING EXPERIENCE

Lecturer, Business Analytics and Technology

Jan 2026 to Present

Universidad Externado de Colombia, Bogota, Colombia

- Teaching the final-semester emphasis of the Business Administration program, covering the full analytics workflow: data quality and preparation, KPI design, applied statistics, predictive modeling (regression and classification), and data ethics.
- Supervising paired student consulting projects end to end, from business problem definition and data sourcing through a Power BI dashboard, executive report and oral defense.

Lecturer, Optimization Methods II

Jan 2026 to Present

Universidad Externado de Colombia, Bogota, Colombia

- Teaching two undergraduate sections covering linear and nonlinear programming, modeling and solution techniques.
- Designing problem sets and assessments; mentoring students in formulating and solving optimization models.

Corporate Instructor, Experimentation and Optimization

Aug 2025 to Mar 2026

Claro Colombia (analytics leadership)

- Designed and delivered applied courses on experimentation and optimization for Claro Colombia's analytics leadership team.

Capstone Research Project, Air Cargo Optimization

Aug 2025 to Feb 2026

Massachusetts Institute of Technology (MIT), Cambridge, MA, USA

- Formulated the Air Cargo Loading Problem as an online 3D bin packing optimization model under real operational constraints.
- Adapted the DeepPack3D framework to heterogeneous ULD configurations of a Boeing 747-400F and implemented heuristic packing strategies (Bottom-Left, Best Volume Fit, Best Long Side Fit).
- Achieved about 90% average volumetric utilization on 16 charter flights, improving the historical baseline (84%) by up to 12 percentage points; estimated economic and CO2 impact.

Teaching Assistant, Logistics and Supply Chain Management

Jul 2023 to Dec 2025

Universidad de La Sabana, Bogota, Colombia

- Supported an undergraduate Industrial Engineering course: leading sessions, guiding exercises and evaluating student work.

Graduate Research Assistant

Jun 2023 to Dec 2025

Universidad de La Sabana, Bogota, Colombia

- Developed Mixed Integer Nonlinear Programming (MINLP) models for competitive and public facility location problems.
- Integrated statistical and multi-criteria decision methods, including discrete choice (random utility) modeling and Fuzzy Analytic Hierarchy Process (Fuzzy AHP).
- Implemented and benchmarked models in Gurobi, GAMS and Python; performed solution validation and runtime analysis.

PROFESSIONAL EXPERIENCE

Senior Data Scientist (Consultant)

Sep 2025 to May 2026

Overlap, Remote, Colombia

- Directed and supervised machine learning projects end to end, from problem framing and modeling to deployment and stakeholder delivery, for clients including BBVA, BCP, Alicorp, Primax, Banco Agrario and Banco Santander.

Senior Data Scientist, Growth & Media

Mar 2025 to May 2026

XAQ10, Remote, USA

- Built measurement, attribution and experimentation frameworks; applied advanced segmentation and cohort analysis.

Growth Specialist (Consultant)

Feb 2025 to May 2025

Dataico, Remote, Colombia

- Designed and deployed machine learning models for customer churn prediction; built automated Python data pipelines and dashboards.

Marketing Data Analyst

Sep 2023 to Oct 2024

Tradeview Markets, Remote

- Built performance dashboards and attribution frameworks across channels; ran audience segmentation and behavioral analysis to improve campaign efficiency.

Optimization & Continuous Improvement Analyst

Dec 2022 to Jun 2023

Coca-Cola FEMSA, Tocancipa, Colombia

- Designed and implemented a large-scale Mixed Integer Programming workforce scheduling model integrating production capacity, labor constraints and demand variability, moving planning from heuristic allocation to mathematical optimization.
- Led Lean Six Sigma continuous improvement projects and built data-driven decision tools for production management.

AWARDS & FELLOWSHIPS

- Graduate Research Assistant Fellowship (100%), Universidad de La Sabana, 2023.
- Best Practitioner Award, Universidad de La Sabana, 2023.
- First Place, Noel x Universidad de La Sabana Innovation Challenge, 2023.
- Top National State Exam Scores in Industrial Engineering, 2023.
- First Place, ASME Impromptu Vehicle Design Competition, 2021.
- Third Place (Latin America), ASME Management Skills Competition, 2021.

PROFESSIONAL MEMBERSHIPS

- The OR Society (United Kingdom).
- Colombian Association of Operations Research (ASOCIO).